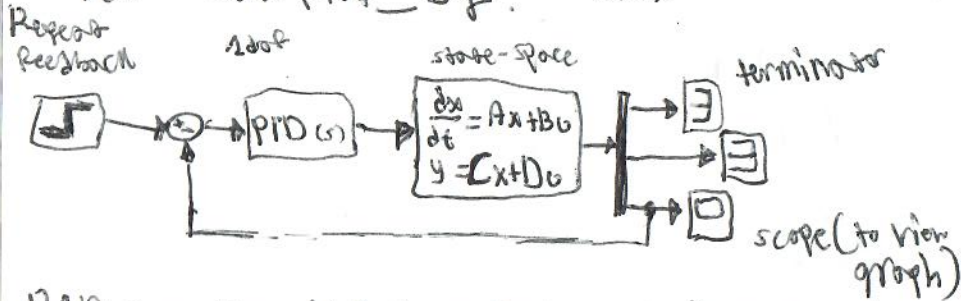


Test Autopilot sys. Demux



PID controller 1 dof : smooths/control aircraft motion by reducing errors & rejecting disturbances

$$P + I \frac{1}{s} + D \frac{N}{1 + N \frac{1}{s}}$$

Proportional (P)

Integral (I)

Derivative (D)

Filter coefficient (N)

State Space: describes how the sys. (aircraft, motor, etc) moves. the letters are the matrices that make that description precise.

A state is set of vars that describes the instantaneous condition of a dynamic sys. so you can predict the future evolution given future inputs.

Integration - Based Project 1

* W = vertical velocity
 U = horizontal velocity

1st coefficient . $a = \tan^{-1}(\frac{W}{U})$

~~4th coefficient~~

~~4th~~

$$CL = 2\pi(\alpha) \rightarrow CL \propto$$

Drag coefficient

$$CD = CD_0 + K(CL)^2$$